

Network Traffic Analysis for DDOS Attack Detection

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ABSTRACT

Distributed Denial of Service attacks (DDoS) are one of the most prevalent attacks threatening systems and their security. In this paper, various models to categorize these attacks are presented, analyzed and compared on regards of their effectiveness for DDoS detection. Machine learning (ML) algorithms for classification are used after pre-processing DDoS dataset to classify network traffic. After analyzing the results of Naïve bayes, Decision Tree, Support Vector Machine, and Random Forest classifiers, we conclude that the most accurate results appeared when using the Random Forest classifier.

CCS CONCEPTS

• Security and Privacy • Intrusion/anomaly detection and malware mitigation • Intrusion detection systems

KEYWORDS

Datasets, Machine Learning, Denial-of-service attack, classifying, Network traffic

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1 Introduction

Distributed Denial of Service attacks are one of the most alarming security threats nowadays. According to Kaspersky's recent report, these attacks are increasing drastically and have doubled in 2019 compared to 2018 [1]. We occasionally hear of “attackers

dropping” a website, and oftentimes, the primary reason behind this failure is the DDoS attack.

A DDoS attack prevents regular traffic from reaching its desired destination by flooding it with unwanted traffic, such as the traffic jam blocking the highway. To launch a DDoS attack on a server or a network, the attacker controlling the bots will give the order to flood the victim's IP with a stream of packets to cause it to overflow, which results in denial of services to normal users [2].

In this paper, a model to accurately detect DDOS attacks using ML algorithms is presented. First by processing the dataset to categorize the network traffic, then building models with various algorithms. Finally, choosing the best model after analyzing and comparing the classification results of different algorithms.

2 Previous studies

Distributed Denial of Service attack is one of the security threats that may cause significant financial losses to organization networks [3]. In this attack, the victim's system will not respond to legitimate traffic due to downtime caused by the large number of requests made by controlled bots. One way for making these requests is by sending the first Syn packet of the TCP three-way handshake and leave the connection open by not completing the handshake. If the downtime lingers, then this may not only cause financial losses, but also may cause irreparable damages to the affected network such as loosing key clients or reputation. Due to the severity of DDoS consequences, many security researchers and data scientists have focused their work on detecting these attacks by analyzing network traffic. By using ML algorithms and malicious/normal datasets to build detection models, accuracy for these models will be high if the suitable algorithms are used [4]. Previous work done on classifying network traffic using ML can be categorized as follows:

- Classification based on the network flow's statistical properties [5]
- Smart detection of DDoS [6]
- Traffic classification using security analysis [7]

The area of network traffic classification use either unsupervised or supervised learning algorithms to detect any traffic pattern.

Authors of [5] proposed Deep Packet Inspection through intrusion detection system device, that should periodically be updated with new traffic patterns to filter malicious traffic. Furthermore, Deep Packet Inspection (DPI) extracts unidirectional and bidirectional flows from a traffic capture and selects statistical properties then uses an unsupervised learning mechanism to cluster flows into groups based on their relationships. The accuracy levels of statistical properties of the traffic flowing in the network for unidirectional flows accuracy is 85% and for bidirectional flows is 86%. However, the proposed system contains network equipment in DPI which needs to be maintained periodically and the maintenance for this equipment can be hard and expensive.

In [6], smart detection of DDoS attacks was proposed that will detect both high and low volume of DDoS attacks. The smart detection acts as a sensor that can be installed anywhere on the network, that classifies online traffic using an ML algorithms-based strategy. The proposed system has higher accuracy than statistical properties and low false positive rate. Nonetheless, the system presented some detection failures.

Traffic loads on the Internet vary from time to time which leads to inaccurate flow feature selection and results in imbalanced traffic flows. Furthermore, imbalance plays a significant part in security analysis and in identification of malicious traffic. Hence, in [7] the authors have proposed a solution for the imbalance by using security analysis with two different metrics for feature selection. These two metrics are Weighted Mutual Information (WMI) and Area Under ROC curve (AUC). WMI is a well-known method that states relationship between two packets. Both metrics support the selection of a range of effective features in imbalanced traffic classification. The proposed system applied ML algorithms to achieve more than 80% flow classification accuracy using different network environment datasets. However, the system has low accuracy results for TCP traffic.

3 Implementation

3.1 Methodology

In our work, different methods will be explained for data processing and data exploratory before implementing the classification approach. After that, various classification algorithms will be applied to train and to select the best model for our project. The imbalanced dataset is used from (CICDDoS2019) [8] dataset, which contains the flow packet information and the label for each flow whether it is DDoS or Benign. Table 1 below represents the general information about imbalanced dataset (CICDDoS2019). Additionally, the number of records within the

dataset will affect the outcomes of the classification models, thus we perform data balancing for the selected dataset.

Table 1: Imbalanced Dataset (CICDDoS2019) General Information

Number of Features	Number of Records	Number of DDoS Records	Number of Benign Records
84	7,616,509	1,523,302	6,093,207

3.2 Data Processing

JupyterLab environment in Anaconda platform is used with Python language, and these libraries were imported Pandas, NumPy, and Plotly to perform data analysis and to support high-level mathematical functions as well as to plot interactive graphs. A combination of libraries was used to perform data processing and to build the accurate detection algorithms. The dataset used is considered huge, which contains 7,616,509 total data points and may result in heavy processing time to run the code. So, stratified sampling is used to take a sample of data that has the same features of the original dataset. Moreover, the following methods are performed to preprocess the dataset:

- **Feature Engineering**

To choose the important features for our model, features will be added or removed after reading the features within the dataset in our environment to reach better accuracy of the model. The preprocessing starts by first, dropping an Unnamed column in Python because it is an index column. Second, all data types of features are identified by built-in functions in Python.

Third, we identified the ratio of DDoS and Benign packets in the dataset. We found that the Benign packets form 83% and DDoS packets form 16% of the entire traffic. After that, Source and Destination Port types were validated. The data type in the dataset for those two features was integer so it means a continuous number, and for modeling purposes it should be converted to a category. Finally, we counted how many data are there within each Source and Destination ports, table 2 shows the sample output of port categorization.

Table 2: Port Categorization

Feature	Port Number	Count
Src Port	80	8881
	443	7489
	445	2153
Dst Port	53	28318
	80	19201
	443	14161

Additionally, a new feature for Source and Destination ports is added which is the port name. The port name is categorized based on Internet Assigned Numbers Authority. The categories are well-known ports, registered ports, and dynamic ports. Each of these

categories has specific numbers and the percentage for these categories in the dataset are calculated. Table 3 shows the percentage of various ports categories in the dataset.

Table 3: Percentage of Port Names

Feature	Well-Known Ports	Registered Ports	Dynamic Ports
Src Port Name	20%	8%	70%
Dst Port Name	68%	16%	15%

Other features are added, which are port length with the respective percentage for each port as shown in table 3 and the port length with the percentage shown in table 4. These features were added to aid building more accurate models. For example, the length of source port 443 is three digits and the percentage of this port length is 10%. After running this feature selection approach, we found out that 77% of the data features in dataset were coming from ports with five digits length. To select best features for the classification models, we followed an approach where we build the models after adding or dropping features, then comparing the accuracy of the built models after each addition or dropping. By doing this, we identified the features that produce the best prediction models.

Table 4: Percentage of Port Length

Feature	Port Length	Percentage
Src Port Len	5	77%
	3	10%
	2	9%
	4	2%
	1	2%
Dst Port Len	2	48%
	5	19%
	3	19%
	4	12%
	1	2%

Finally, we added a feature which is the protocol name with its percentage. After adding this feature, we found that TCP formed 69%, TDP formed 29%, and HOPOPT formed only 1% of the total traffic.

- **Missing Value treatment**

This treatment is performed by checking if any column in the dataset is Null. There were missing values in Flow Bytes/s column, so the column is dropped as it will reduce the power of the model. This treatment can be performed by One-Hot Encoding. With this method, categorical values are represented as binary to help in building the accurate model. First, categorical values will be mapped to integer values, then each integer value is represented as a binary vector that is all zero values except index of the integer, which is marked with number one.

Before implementing the encoding, some variables are dropped which are Src Port, Dst Port, and Flow ID, since the type of these variables is object. After that, the categorical variables Flow

Bytes/s and Flow Pkts/s needed to be converted into numerical variables. Furthermore, Src Port Len, Dst Port Len, and Protocol variables are converted from numerical to categorical, because each number in these variables represented has a meaning in categorical form of data, and we can keep track of the categories in a numerically meaningful way. Then, unnecessary columns variables with object type are excluded which are Src IP, Dst IP, Timestamp, Protocol, and Label. Finally, the variables that must be converted to One-Hot Encoding are Src Port Name, Dst Port Name, Src Port Len, Dst Port Len, and Protocol Name.

While the One-Hot Encoding is implemented for the selected variables, the number of unique values within each variable were examined, and a flag for dummy variables were added. This may show variables that are highly correlated, as dummy variable trap is used. The first categorical variables should be dropped then check the total number of variables after creating dummy variables by implementing a loop. The first loop was 87 variables and the last loop was 95 variables.

- **Outlier Treatment**

Machine Learning deals with the accurate distribution of data points. For this reason, we must detect points that are outside the overall pattern of distribution. For our purpose, we will get the scatter plot only from the numerical columns thus we detect and remove any other variables such as the number of numerical type variables, flagged variables, and number of numerical variables without flagged variables. The outlier variables are Src Port Name, Dst Port Name, Src Port Len, Dst Port Len, Protocol Name, Src IP, Dst IP, Timestamp, and Protocol by adopting outlier treatment technique [9].

3.3 Exploratory Data Analysis

In this section, we are using exploratory data analysis approach to review and analyze the main features of the whole dataset (CICDDoS2019) [8] to understand how the dataset looks like and to find out which variables are linked together. The analysis with the use of ML algorithms, will make it easier to make a prediction. Listed below are different types of analysis that were performed on the dataset.

- **Scatter Plot**

We generated the scatter plot matrix for the dataset by using Plotly library, the dataset values of Benign and DDOS represented by a dot. The red dots represent DDOS values and the blue dots represent Benign values in each column.

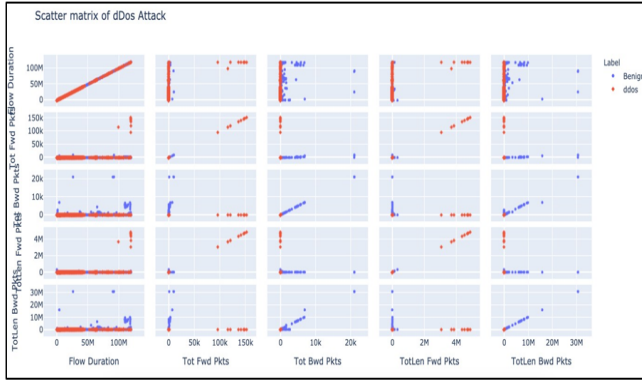


Figure 1: Scatter Matrix

Figure 1 represents the scatter plot matrix with the x-axis and the y-axis representing columns of the dataset to show how the features are linked together. For example, figure 2 shows the relationship between the Flow Duration in x-axis and the Total Forward Packets 'Tot Fwd Pkts' in y-axis. We found that the 85 in Forward Packets are linked to DDOS attack.

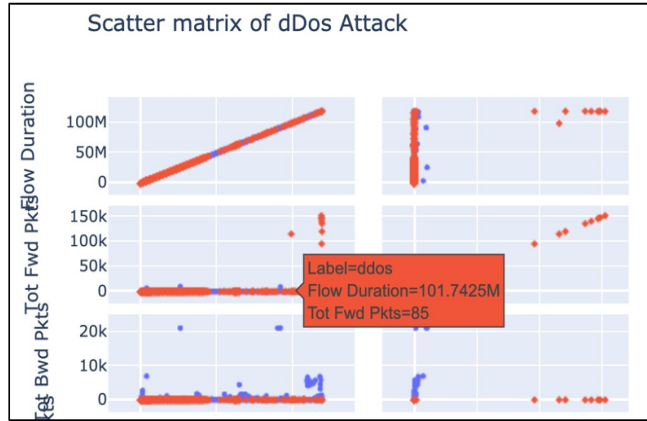


Figure 2: Relationship of Flow Duration and Total Forward Packets

- **Basic Statistic**

Basic statistics help to display spreading of the data in a table format by using Univariate analysis which contains (Mean, Standard Deviation, Minimum, Maximum, and Percentage) of each variable in the dataset. Table 5 is the generated statistics for the dataset.

Table 5: Basic Statistic

Variables	mean	std	min	25%	50%	75%	max
TotLen Fwd Pkts	781.13354	41170.04496930464	0	0	44	621	4892032
TotLen Bwd Pkts	4668.3051	211938.94490961527	0	0	111	488	30881520
Fwd Pkt Len Max	219.38721	325.35898976839843	0	0	42	381	5637
Fwd Pkt Len Min	11.864	24.304837610068187	0	0	0	31	1472
Fwd Pkt Len Mean	57.83248274029957	71.60869988079321	0	0	38	73.58823529411767	1472
Fwd Pkt Len Std	82.29598076361344	138.54391114490363	0	0	0	113.4404950668525	1968.3534589044946
Bwd Pkt Len Max	354.53667	512.7864724373577	0	0	99	409	5840
Bwd Pkt Len Min	28.438	52.413541645240194	0	0	0	52	2065
Bwd Pkt Len Mean	114.10370266254846	166.2697720959343	0	0	71	150	2291.606156501717
Bwd Pkt Len Std	130.06218284986195	205.15108312685496	0	0	0	188.68643679017163	1700.8383586402979
Flow Bw/s	null	null	0	0	916.2454792999999	82263.18087015569	null
Flow Pkts/s	null	null	0.01667132519063577	4.883870719207694	112.3674208803472	5344.531983508185	null
Flow IAT Mean	2288093.5207210807	10643383.94563477	0	345	13603.333333333334	227487.82996742666	119966468
Flow IAT Std	1125375.5934233046	3879684.60587114	0	0	109.60155108391487	250291.6226827994	84568020.12230179
Flow IAT Max	5357671.37756	15371543.633636778	0	417	24496	1120000.5	119966468
Flow IAT Min	1792952.62595	10553959.480553104	-1	5	50	940	119966468
Fwd IAT Tot	11145679.58259	30566779.777875222	0	0	804	1637297.75	120000000
Fwd IAT Mean	2548474.905917006	10812394.511533497	0	0	398	243871.46428571426	119966468

- **Percentile Distribution**

Percentile distribution is performed for each variable in the dataset to provide numbers where certain percentages of scores fall below these numbers. Table 6 below shows the percentile distribution.

Table 6: Percentile Distribution

Variables	Min	Percentile 10	Percentile 25	Percentile 50	Percentile 75	Percentile 85	Percentile 90	Percentile 95	Percentile 99	Max
Flow Duration	0	77	549	28945.5	3116420.75	5996199.249999	54330731.40000	112876918.0499	118200688.8399	119999020
Tot Fwd Pkts	0	1	1	2	5	9	10	16	36	152876
Tot Bwd Pkts	0	0	1	1	5	7	9	14	34	21209
TotLen Fwd Pkts	0	0	0	44	621	1032	1144	1164	2466.009999999	4892032
TotLen Bwd Pkts	0	0	0	111	488	1581	1731	4472	21346.149999999	30881520
Fwd Pkt Len Max	0	0	0	42	381	677	677	935	993	5637
Fwd Pkt Len Min	0	0	0	0	31	39	42	46	64	1472
Fwd Pkt Len Mean	0	0	0	38	73.58823529411	129	143.5	233.75	233.75	1472
Fwd Pkt Len Std	0	0	0	0	113.4404950666	218.5088353159	228.1296623788	467.5000000000	467.5000000000	1968.353458904
Bwd Pkt Len	0	0	0	99	409	1173	1173	1460	1460	5840

- **Mean Bar Plot for All Numerical Variables**

The mean of numerical data within the dataset will be represented in Bar graph based on five criteria: Source and Destination port name, Protocol name, Source and Destination port length. Furthermore, the graph shows whether the traffic is DDOS or benign for each of the criteria. For example, the graphs in figure 3 for "Fwd Seg Size Min" feature shows all the criteria and where the DDOS traffic is. We can check the well-known destination port and dynamic source port that have DDOS traffic.

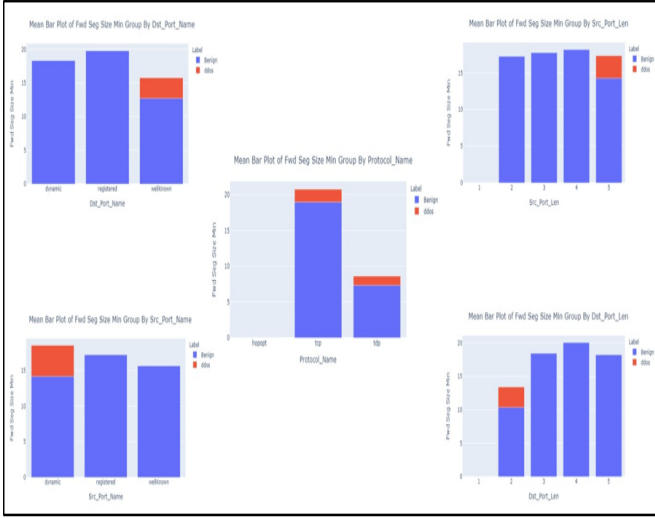


Figure 3: Mean Bar Plot of Fwd Seg Size Min

3.4 Detection Algorithms for Model Building

Before model building, features in training and testing dataset must be normalized. After that, building the required classification algorithms as well evaluate them to find the accurate model based on our findings. Below the steps we take into our consideration to proceed with the model selection.

- **Data Split**

After the complete data analysis, we will split the dataset (CICDDoS2019) [8] into two parts, one part for the training (75%) and the other part is for the testing (25%).

- **Data Normalization**

The features in the training dataset may have odd behavior or the vectors do not unite due to scaling issues. Therefore, we normalize the data to have more consistent features. The data will be populated correctly to get normalized distribution for the data. First, we store the minimum and maximum values of each numerical columns from the training dataset. Second, the formula of normalization

$$z = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Where z is the data will be applied. Then, the normalized data will be represented from 0 to 1. Finally, we normalize the testing dataset following the same steps.

- **Building Classification Algorithms**

To develop the algorithms and to evaluate all models we have three phases:

- Applying algorithms on training dataset and testing dataset.
- Find out the important features within the original dataset.

- Evaluate all models and select the best one based on accuracy, recall, precision, and f1 score.

Normally, we want our model to identify the maximum number of DDOS attacks in the traffic and to differentiate it from normal traffic. To develop any model, we first initiate the model's name, then we create the model from the testing and the training dataset. After that, we check the accuracy, the precision, the recall, and the f1 score in these two datasets [10].

3.5 Model Selection

We have experimented with a number of classification algorithms which are Naïve base, Decision Tree, SVM (Support Vector Machine), and Random Forest. Based on the experimental results and the evaluation criteria shown in figure 4, the best prediction results were achieved when using Random Forest Classifier. Random forest works by creating multiple decision trees and calculating the average of these decision trees.

Figure 5 shows the important features that when used in our model, the prediction accuracy achieved was the best. From the figure 5, out of the 84 features, only 33 features were selected for their importance. These features were selected based on the built Random Forest model as shown in figure 6.

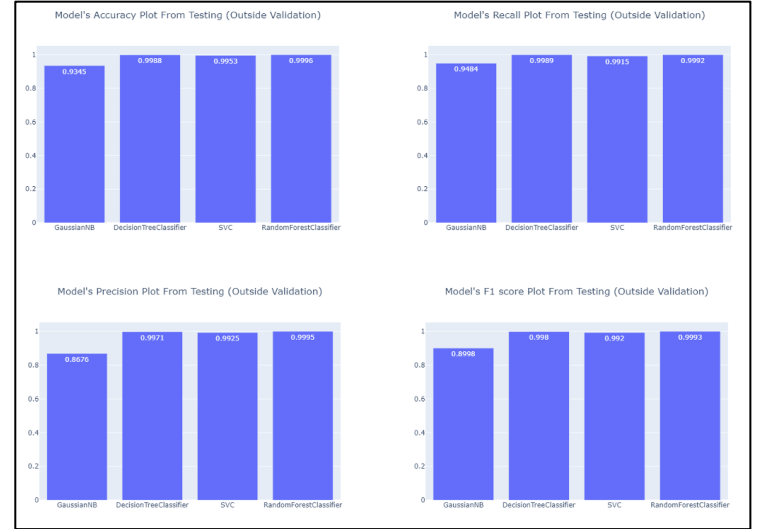


Figure 4: Evaluation of Classification Models

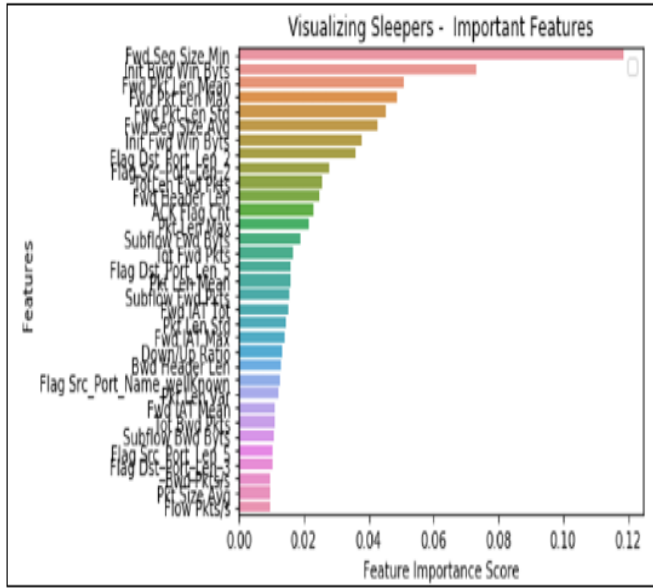


Figure 5: Important Features

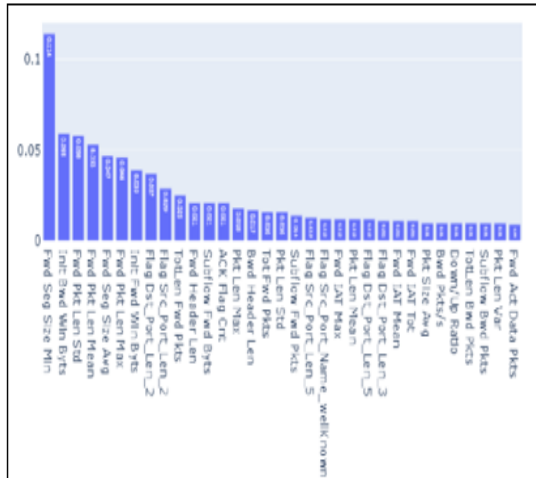


Figure 6: Features Importance Score from Random Forest Plot

4 Conclusion

In this paper, we have presented our work on analyzing network traffic dataset to help enhancing the detection of DDoS attacks. DDoS attacks can cause devastated harm to organizations thus the work on this paper can be beneficial to any network security professional or administrator trying to prevent harmful traffic. Our work focused on improving the classification algorithms used for predicting DDoS attacks by deeply analyzing and understanding the features used for building the classification models. After preprocessing, cleansing, and analyzing these features, we have

come up with the best features for the prediction. Furthermore, we have compared several classification algorithms for the purpose of predicting DDoS and found out that Random Forest gave the best results. The work on this paper can be improved in the future by considering other datasets to check the validity of the findings.

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